

ELEC70122: Machine Learning For Safety-Critical Decision Making

Lecture 4: Learning to Defer in Uncertainty

Dr Sonali Parbhoo

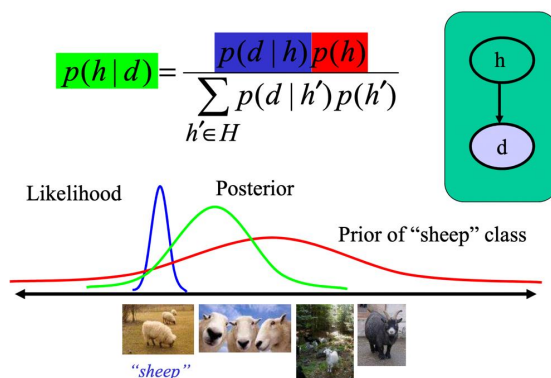
AI for Actionable Impact (AI4AI), EEE

Imperial College London

s.parbhoo@imperial.ac.uk

February 2, 2026

So far



Bayesian Modelling Process

So far

Covariate Shift

$$P_{\text{train}}(X) \neq P_{\text{deploy}}(X)$$

What changes:

- Input distribution shifts
- Environment, population and/or sensors change

What stays the same:

- $P(Y|X)$ assumed to stay the same.

Implications:

- Decisions trained on common cases will fail in rare regimes.

Label Shift

$$P_{\text{train}}(Y) \neq P_{\text{deploy}}(Y)$$

What changes:

- Label distribution/rates of outcomes shift

What stays the same:

- $P(X)$ assumed to stay the same.

Implications:

- Decision thresholds may become poorly calibrated.
- E.g. Disease prevalence/failure rates change across populations.

Concept Shift

$$P_{\text{train}}(Y | X) \neq P_{\text{deploy}}(Y | X)$$

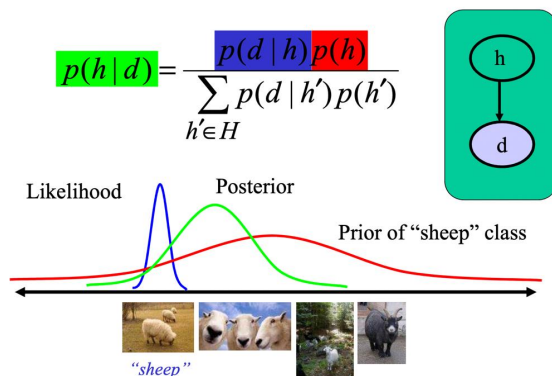
What changes:

- Meanings of features and/or outcomes changes
- Relationships between Y and X evolve

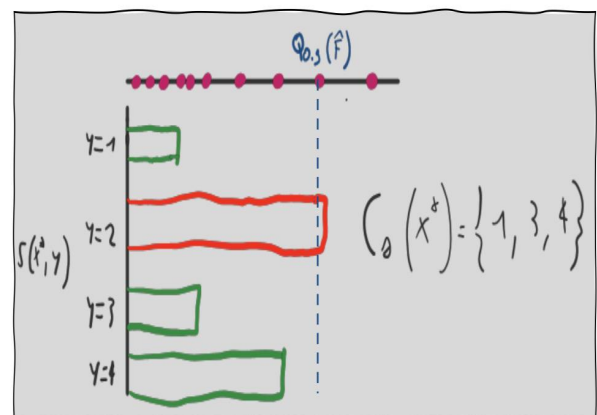
Implications:

- Past data no longer reflects current causal structure

So far



Bayesian Modelling Process



Uncertainty quantification without distributions: Conformal Prediction

What do we do with **uncertainty**?

Outline for today's lecture

- Learning to defer
- Probabilistic learning to defer
- Extensions to sequential settings

What do we do when there is uncertainty or disagreement?

- When there is disagreement between what a doctor versus what a model might suggest, we need to **learn to defer** -> how to combine different models.
- Traditionally modelled as a **mixture-of-experts** model

Mixtures-of-Experts

- Define a variety of “experts” and a gating function
- Gating can be
 - “hard” if it chooses one expert
(e.g., if different experts are good at different domains)
 - “soft” if it combines different experts’ outputs

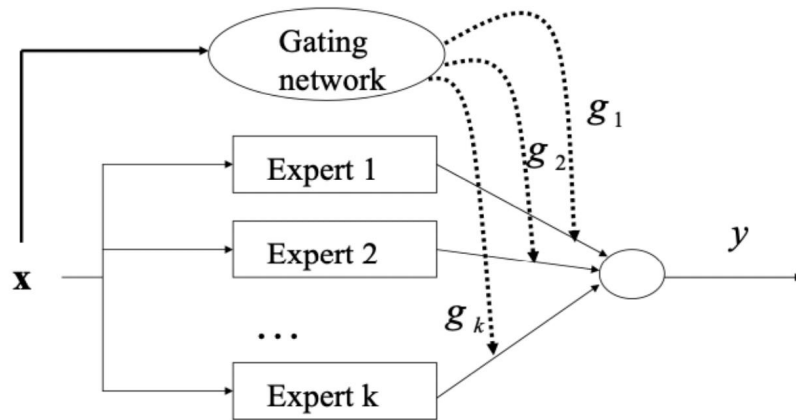


Image from Milos Hauskrecht, <https://people.cs.pitt.edu/~milos/courses/cs2750-Spring04/lectures/class22.pdf>

7

Hierarchical (Soft) Mixtures-of-Experts

- Generalized Linear Models for experts & gating
- Expert network (i, j) produces output $\mu_{ij} = f(U_{ij}\mathbf{x})$ [f is the non-linearity]
- Gating networks compute a linear function and then softmax

$$\xi_i = \mathbf{v}_i^T \mathbf{x}, \text{ and } g_i = e^{\xi_i} / \sum_k e^{\xi_k}$$

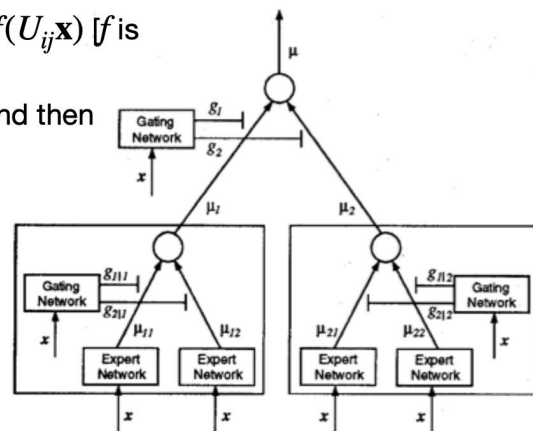
- Lower levels use $g_{j|i}$ and $\xi_{ij} = \mathbf{v}_{ij}^T \mathbf{x}$

$$\mu_i = \sum_j g_{j|i} \mu_{ij}$$

- Probabilistic interpretation:

$$P(y | \mathbf{x}, \theta^0) = \sum_i g_i(\mathbf{x}, \mathbf{v}_i^0) \sum_j g_{j|i}(\mathbf{x}, \mathbf{v}_{ij}^0) P(y | \mathbf{x}, \theta_{ij}^0)$$

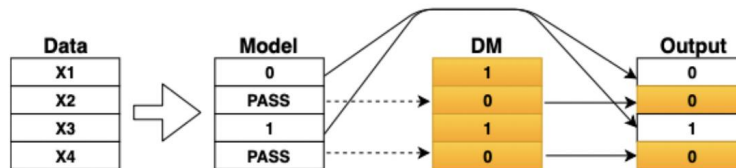
- Used EM to train; lots of hair, as usual!



c.f., Random Forests

Learning-to-Defer

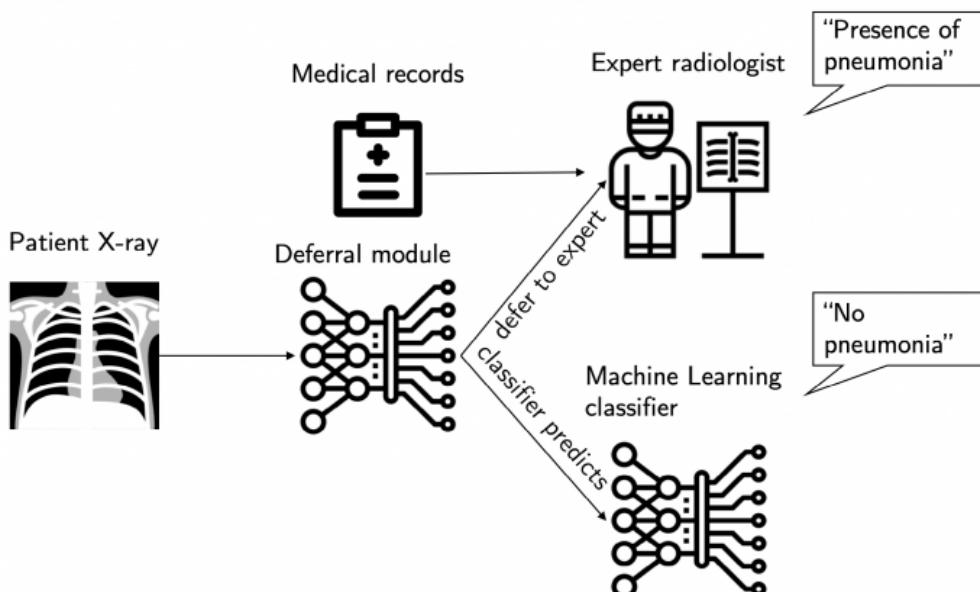
- Assume two “systems”
 - “Model” can either decide (binary choice: 0/1) or **PASS**
 - Decision Maker (DM) may be better, but more costly (e.g., may be human)
 - May have additional inputs
 - Opaque



- Uses of first-stage model:
 - Flag difficult cases for review
 - Cull a large pool of cases
 - Audit DM for bias
 - ...

9

Learning-to-Defer



Qu: How to combine model and expert to 1) achieve better performance and 2) ease the burden on the radiologist?

A: Given only patient input, learn model to route decision to either expert or model in order to maximize system performance

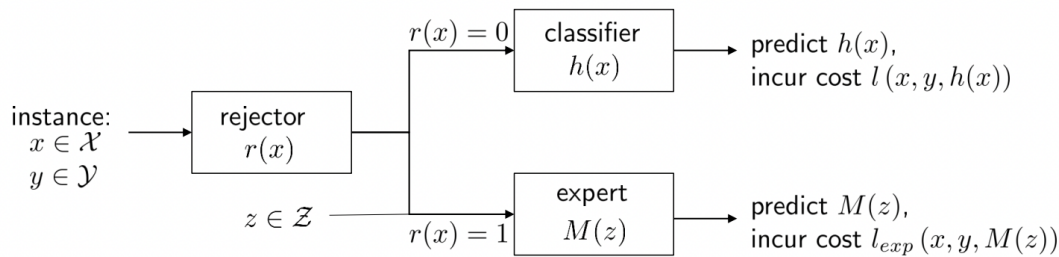
10

Many papers on this....

- Madras et al. (NeurIPS 2018) proposes a MoE loss, resulting loss is not consistent and fails empirically.
- Raghu et al. (2019) propose a confidence score method that compares expert and algorithm confidence. However, classifier cannot adapt to expert.
- Det et al. (AAAI 2020) gives an approximate algorithm for ridge regression, Wilder et al. (IJCAI 2020) combines mixtures of experts loss and confidence score comparison.
- Parbhoo et al (2021) propose preferential MoEs that align with human expertise as much as possible
- Parbhoo et al (AMIA 2016, PloS ONE 2017) propose MoEs for application in HIV therapy selection
- Related problems: selective classification (Geifman & El-Yaniv, NeurIPS 2017), learning with a reject option (Ni et al., NeurIPS 2019)

11

Learning to Defer: Problem Formulation



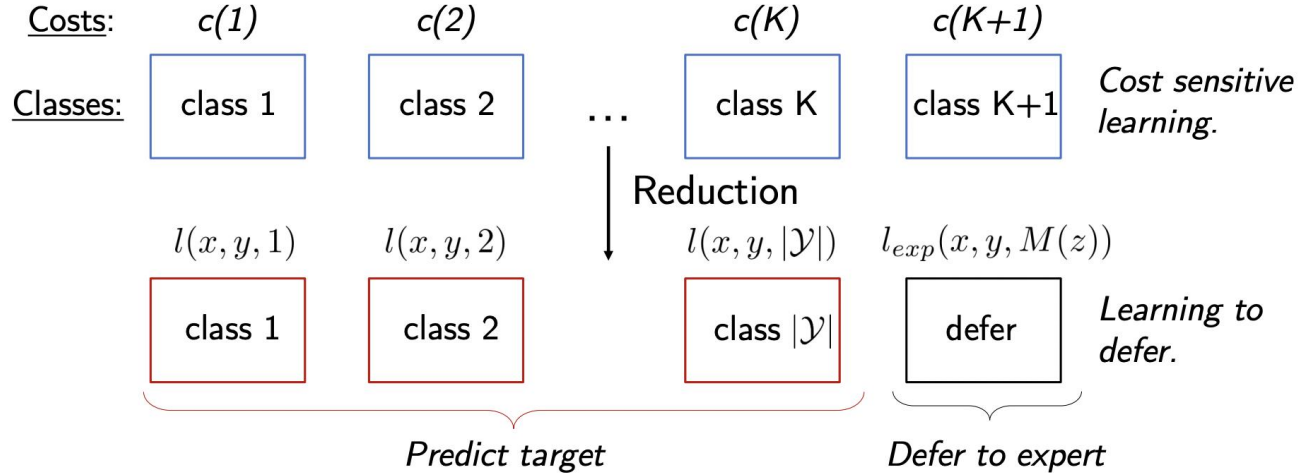
- **Jointly** learn a classifier $h(x)$ and rejector $r(x)$ to minimize system loss:

$$L(h, r) = \mathbb{E}_{(x, y) \sim \mathbf{P}, m \sim M | (x, y)} \left[\underbrace{l(x, y, \hat{Y}(x))}_{\text{classifier cost}} \overbrace{\mathbf{I}_{r(x)=0}}^{\text{predict}} + \underbrace{l_{\text{exp}}(x, y, m)}_{\text{expert cost}} \overbrace{\mathbf{I}_{r(x)=1}}^{\text{defer}} \right]$$

12

Learning to Defer: Reduction to cost-sensitive learning

- Cost sensitive learning: given covariate x pick class in $[K+1]$ that has minimal cost:



Mozannar, Hussein, and David Sontag. "Consistent estimators for learning to defer to an expert." *International conference on machine learning*. PMLR, 2020.

13

Learning to Defer: Reduction to cost-sensitive learning

- Can be formulated as a surrogate function that extends the cross-entropy loss

let $g_i : \mathcal{X} \rightarrow \mathbb{R}$ for $i \in [K + 1]$ and $h(x) = \arg \max_i g_i$, define

$$\begin{aligned} & \tilde{L}_{CE}(g_1, \dots, g_{K+1}, x, c(1), \dots, c(K+1)) \\ &= - \sum_{i=1}^{K+1} \left(\max_{j \in [K+1]} c(j) - c(i) \right) \log \left(\frac{\exp(g_i(x))}{\sum_k \exp(g_k(x))} \right) \end{aligned}$$

- Corresponds to multiclass misclassification error.

Mozannar, Hussein, and David Sontag. "Consistent estimators for learning to defer to an expert." *International conference on machine learning*. PMLR, 2020.

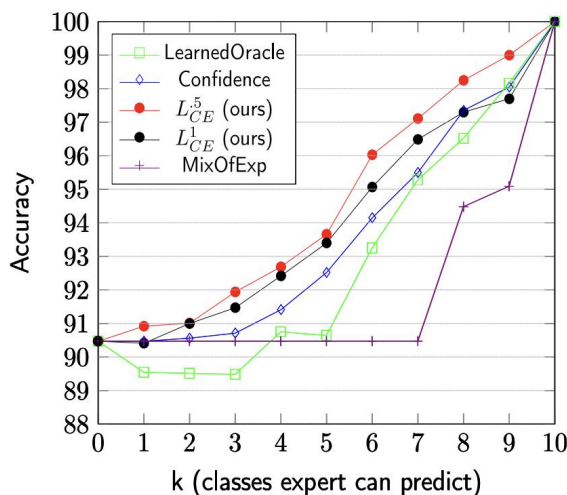
14

Experiments

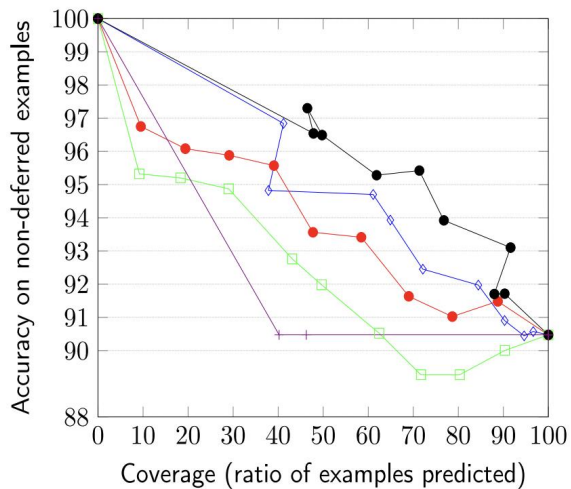
- **CIFAR 10:** image classification over 10 classes, parameterize model g as a WideResNet with 11 output layers, no data augmentations were used.
- **Synthetic expert:** let $1 \leq k \leq 10$, then if the image belongs to the first k classes the expert predicts perfectly, otherwise the expert predicts uniformly at random.
- **Baselines:** 1) MixOfExp (Madras et al. 2018), 2) Confidence (Raghu et al. 2019), 3) LearnedOracle: build model to predict if image is in first k classes and defer accordingly

15

Results



Accuracy of combined system for each expert k

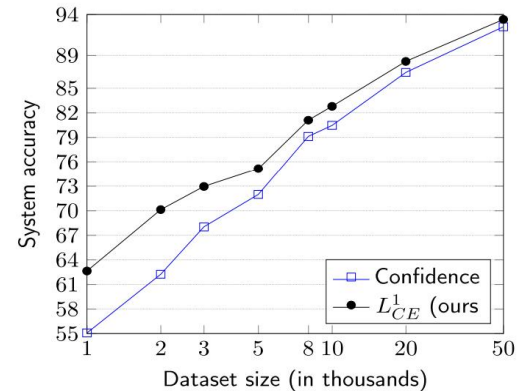


Accuracy of classifier and its coverage for each expert k

16

Why does it work?

1. **Sample Complexity:** as we restrict training data, gains over Confidence increase
2. **Considering classifier's confidence:** LearnedOracle baseline does not look at confidence of classifier and hence suffers.
3. **Consistency:** MixOfExp baseline is not consistent, there is a mismatch between the loss and actual misclassification error



Restricting training data size and showing system accuracy

Mozannar, Hussein, and David Sontag. "Consistent estimators for learning to defer to an expert." *International conference on machine learning*. PMLR, 2020.

17

Real Experiments

- **CheXpert:** large chest X-ray dataset with over 224k automatically labeled images for the presence of 14 observations (Irvin et al., 2019)
- **Synthetic expert:** if patient has supporting device, expert is correct with probability p , otherwise expert is correct with probability q
- **Baselines:** 1) Confidence (Raghu et al., 2019), 2) ModelConfidence: defer based on confidence of model
- **Task:** We constrain our method and the baselines to achieve $c\%$ coverage and measure AU-ROC & AU-PR of the system.

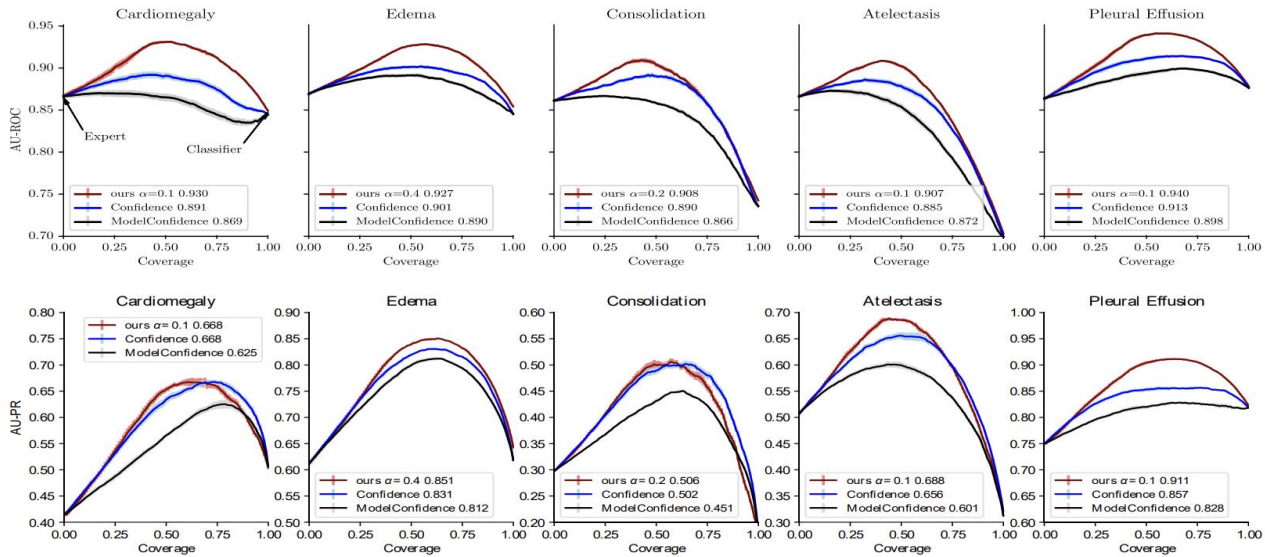


Chest X-ray of patient with Cardiomegaly

Mozannar, Hussein, and David Sontag. "Consistent estimators for learning to defer to an expert." *International conference on machine learning*. PMLR, 2020.

18

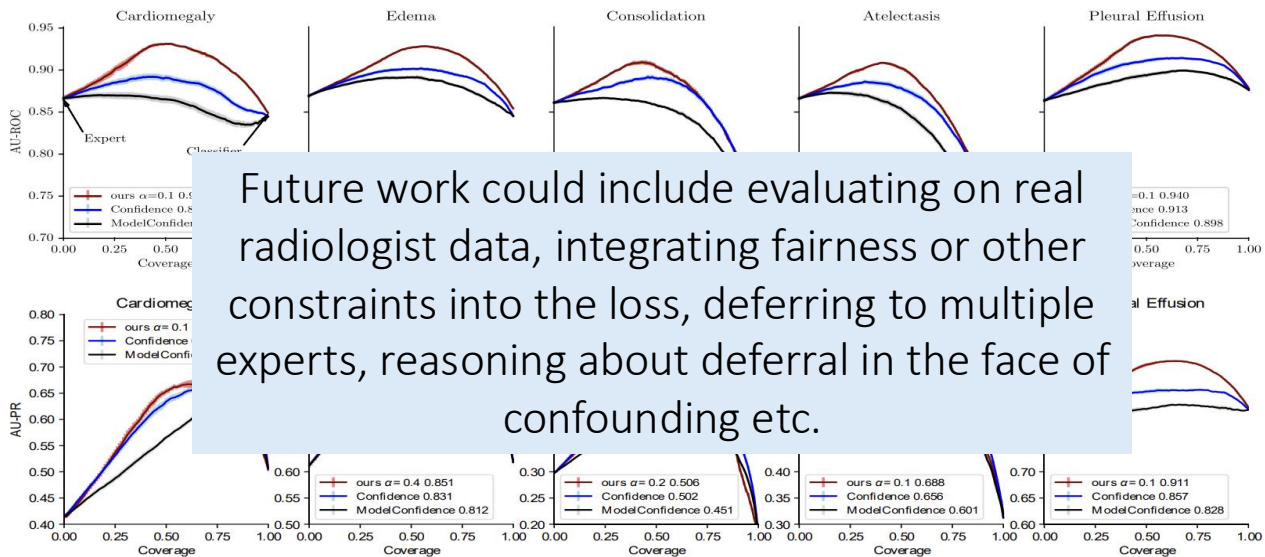
Results



Plot of AU-ROC of the ROC curve (a) for each level of coverage and of the AU-PR (AP) (b) for each of the 5 tasks comparing our method with the baselines on the training derived test set for the toy expert with $q=0.7$, $p=1$.

19

Results

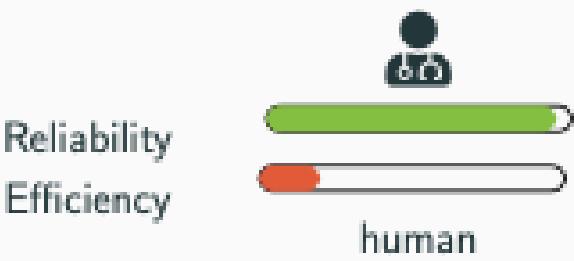


Future work could include evaluating on real radiologist data, integrating fairness or other constraints into the loss, deferring to multiple experts, reasoning about deferral in the face of confounding etc.

Plot of AU-ROC of the ROC curve (a) for each level of coverage and of the AU-PR (AP) (b) for each of the 5 tasks comparing our method with the baselines on the training derived test set for the toy expert with $q=0.7$, $p=1$.

20

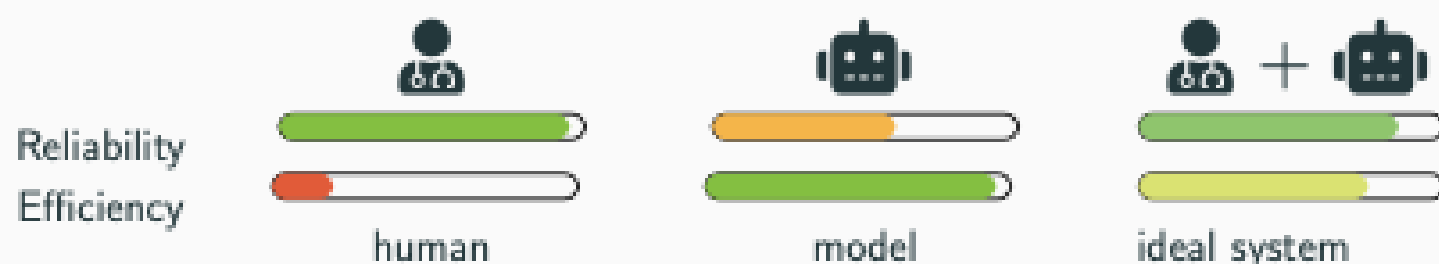
Introduction



Introduction

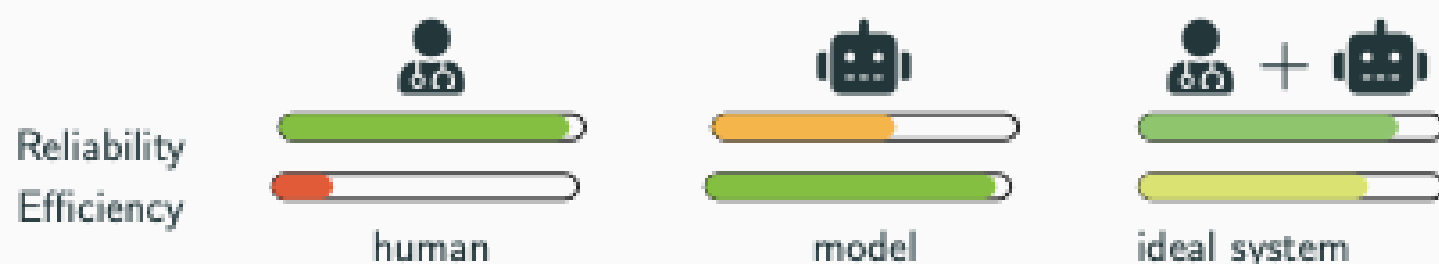


Introduction



2

Introduction



Learning to defer (L2D) aims to leverage:

- high *reliability* of human, and
- high *efficiency* of machine learning models.

2

Background - Learning to defer

L2D can be seen under 2 different models:

- **Mixture of experts** consists of 2 phases: expert selection and prediction,
- “Unified” L2D outputs both expert selection and classifier’s prediction.

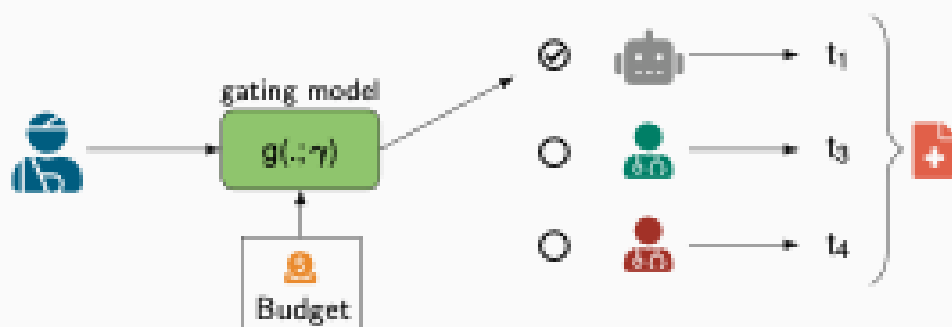
3

Background - Learning to defer

Mixture of experts¹

The model consists of:

- a gating model,
- M human experts, and
- a classifier.

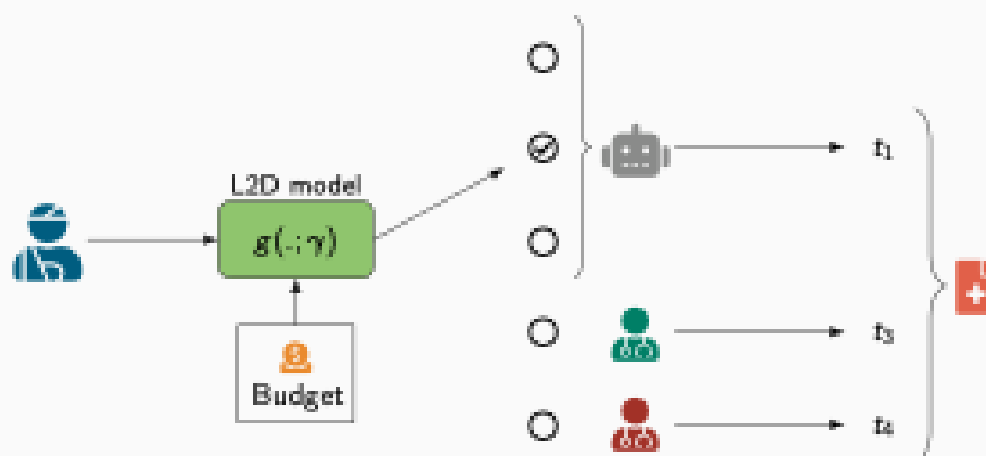


¹David Madras, Toni Pitassi, and Richard Zemel. “Predict responsibly: Improving fairness and accuracy by learning to defer”. In: *Advances in Neural Information Processing Systems*. 2018.

4

Background - Learning to defer

“Unified” L2D² integrates both the classifier’s prediction and expert selection into a single model.



²Hussein Mozannar and David Sontag. “Consistent estimators for learning to defer to an expert”. In: *International Conference on Machine Learning*. PMLR, 2020, pp. 7076–7087.

Limitations of current L2D

			...	
X_1	$t_1^{(1)}$	$t_1^{(2)}$	...	$t_1^{(M)}$
X_2	$t_2^{(1)}$	$t_2^{(2)}$	...	$t_2^{(M)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	$t_N^{(1)}$	$t_N^{(2)}$	...	$t_N^{(M)}$
Complete annotation				

⚠ requires *all* human experts must annotate every training sample

Limitations of current L2D

			...	
X_1	$t_1^{(1)}$	$t_1^{(2)}$	...	$t_1^{(M)}$
X_2	$t_2^{(1)}$	$t_2^{(2)}$	...	$t_2^{(M)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	$t_N^{(1)}$	$t_N^{(2)}$	...	$t_N^{(M)}$
Complete annotation				

→

			...	
X_1	×	$t_1^{(2)}$	...	×
X_2	$t_2^{(1)}$	×	...	×
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	×	×	...	$t_N^{(M)}$
In-complete annotation				

⚠ requires *all* human experts must annotate every training sample

6

Limitations of current L2D

			...	
X_1	$t_1^{(1)}$	$t_1^{(2)}$	...	$t_1^{(M)}$
X_2	$t_2^{(1)}$	$t_2^{(2)}$	...	$t_2^{(M)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	$t_N^{(1)}$	$t_N^{(2)}$	...	$t_N^{(M)}$
Complete annotation				

→

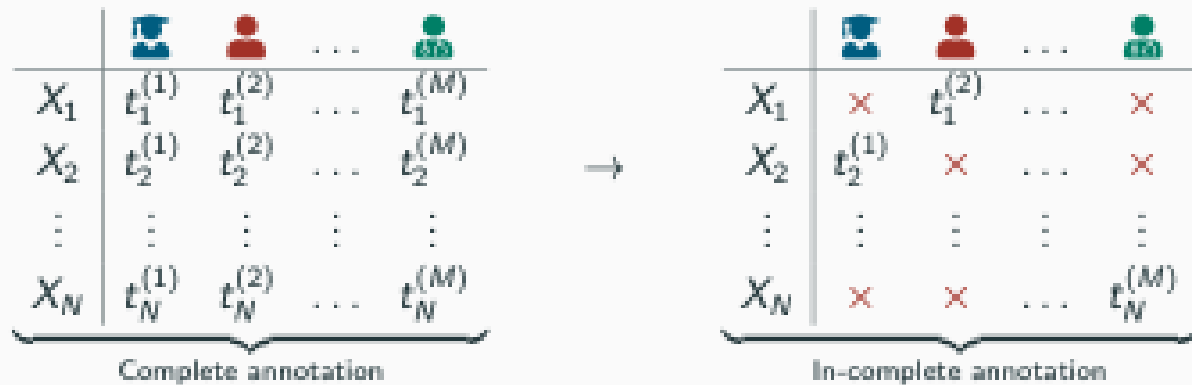
			...	
X_1	×	$t_1^{(2)}$	...	×
X_2	$t_2^{(1)}$	×	...	×
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	×	×	...	$t_N^{(M)}$
In-complete annotation				

⚠ requires *all* human experts must annotate every training sample

🚫 impractical (e.g., each sample is annotated by few human experts), and

6

Limitations of current L2D



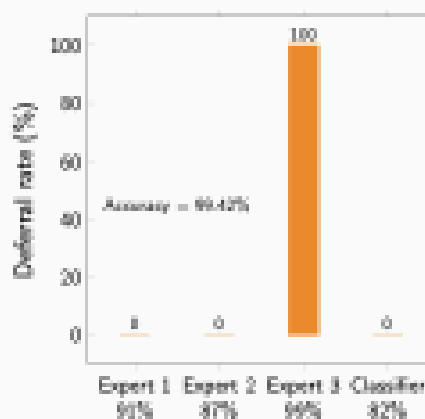
- ⚠ requires *all* human experts must annotate every training sample
- 🚫 impractical (e.g., each sample is annotated by few human experts), and
- 💰 costly, time-consuming, and even infeasible (e.g., radiology³),

³Leonard Berlin. "Liability of interpreting too many radiographs". In: *American Journal of Roentgenology* 175.1 (2000), pp. 17–22.

6

Limitations of current L2D (cont.)

- ⚠ most likely selects the *best human expert* all the time
- 👤 unfair workload assignment, and
- 😞 fatigue, burnout → misdiagnosis.

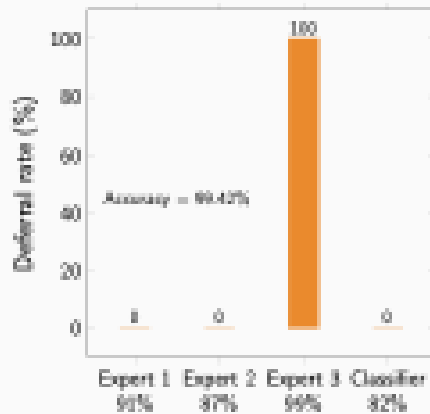


(a) Imbalanced workload

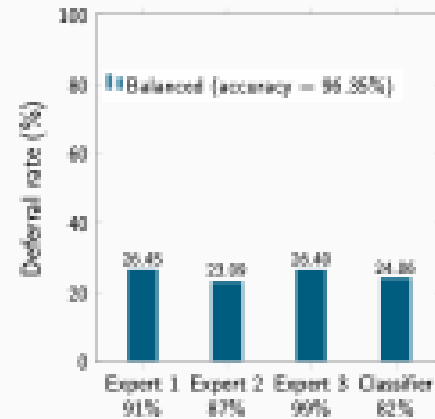
7

Limitations of current L2D (cont.)

- ⚠️ most likely selects the *best human expert* all the time
- 👤 unfair workload assignment, and
- 😫 fatigue, burnout → misdiagnosis.



(a) Imbalanced workload



(b) Balanced workload

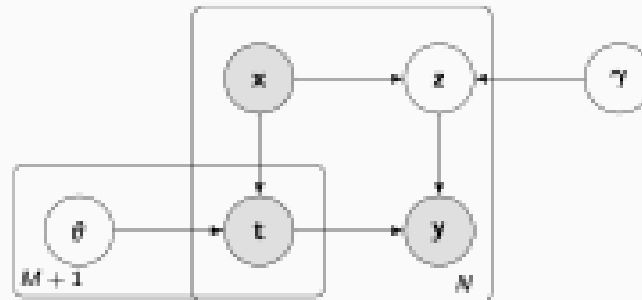
Probabilistic L2D - Complete annotations

Probabilistic L2D is based on the *mixture of experts* modelling approach.

Probabilistic L2D - Complete annotations

Probabilistic L2D is based on the *mixture of experts* modelling approach.

With “complete” annotation data, L2D can be modelled as a graphical model:

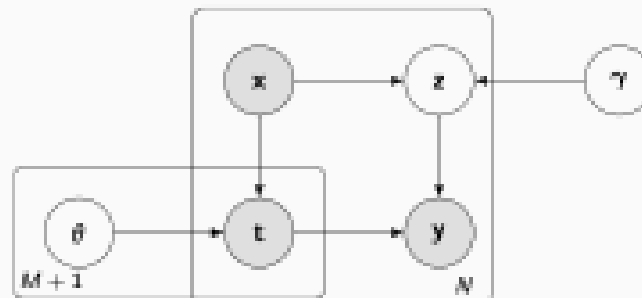


1. draw a sample: $\mathbf{x} \sim \Pr(\mathbf{x})$,
2. draw expert's annotation: $\mathbf{t}^{(m)} \sim \Pr(\mathbf{t}|\mathbf{x}, \theta_m) = \text{Categorical}(\mathbf{t}|f(\mathbf{x}; \theta_m))$,

Probabilistic L2D - Complete annotations

Probabilistic L2D is based on the *mixture of experts* modelling approach.

With “complete” annotation data, L2D can be modelled as a graphical model:

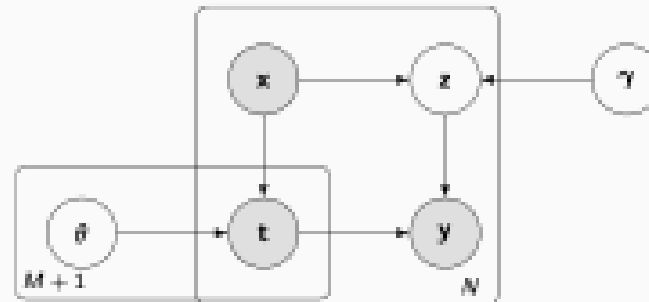


1. draw a sample: $\mathbf{x} \sim \Pr(\mathbf{x})$,
2. draw expert's annotation: $\mathbf{t}^{(m)} \sim \Pr(\mathbf{t}|\mathbf{x}, \theta_m) = \text{Categorical}(\mathbf{t}|f(\mathbf{x}; \theta_m))$,
3. draw an expert: $\mathbf{z} \sim \Pr(\mathbf{z}|\mathbf{x}, \gamma) = \text{Categorical}(\mathbf{z}|g(\mathbf{x}; \gamma))$,

Probabilistic L2D - Complete annotations

Probabilistic L2D is based on the *mixture of experts* modelling approach.

With “complete” annotation data, L2D can be modelled as a graphical model:



1. draw a sample: $\mathbf{x} \sim \Pr(\mathbf{x})$,
2. draw expert's annotation: $\mathbf{t}^{(m)} \sim \Pr(\mathbf{t}|\mathbf{x}, \theta_m) = \text{Categorical}(\mathbf{t}|f(\mathbf{x}; \theta_m))$,
3. draw an expert: $\mathbf{z} \sim \Pr(\mathbf{z}|\mathbf{x}, \gamma) = \text{Categorical}(\mathbf{z}|g(\mathbf{x}; \gamma))$,
4. draw the ground truth: $\mathbf{y} \sim \Pr(\mathbf{y}|\mathbf{z}, \mathbf{t}) = \text{Categorical}(\mathbf{y}|\mathbf{t}^{(z)})$,

8

Probabilistic L2D - Missing annotations

Probabilistic L2D - Missing annotations

Relax the assumption of annotation availability

			...	
X_1	$t_1^{(1)}$	$t_1^{(2)}$	...	$t_1^{(M)}$
X_2	$t_2^{(1)}$	$t_2^{(2)}$	...	$t_2^{(M)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	$t_N^{(1)}$	$t_N^{(2)}$	...	$t_N^{(M)}$
Complete annotation				

→

			...	
X_1	×	$t_1^{(2)}$	...	×
X_2	$t_2^{(1)}$	×	...	×
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	×	×	...	$t_N^{(M)}$
In-complete annotation				

Probabilistic L2D - Missing annotations

Relax the assumption of annotation availability

			...	
X_1	$t_1^{(1)}$	$t_1^{(2)}$	...	$t_1^{(M)}$
X_2	$t_2^{(1)}$	$t_2^{(2)}$	...	$t_2^{(M)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	$t_N^{(1)}$	$t_N^{(2)}$	...	$t_N^{(M)}$
Complete annotation				

→

			...	
X_1	×	$t_1^{(2)}$	...	×
X_2	$t_2^{(1)}$	×	...	×
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_N	×	×	...	$t_N^{(M)}$
In-complete annotation				

$t_i^{(m)} \forall m \in \{1, \dots, M\}$ is observed. → Some $t_i^{(j)}$ are missing (e.g., latent).

Probabilistic L2D - Missing annotations

Objective

$$\max_{\gamma, \{\theta_m\}_{m=1}^{M+1}} \sum_{i=1}^N \ln \Pr \left(\mathbf{y}_i, \prod_{m \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_j^{(m)} \middle| \mathbf{x}_i, \gamma, \{\theta_m\}_{m=1}^{M+1} \right),$$

10

Probabilistic L2D - Missing annotations

Objective

$$\max_{\gamma, \{\theta_m\}_{m=1}^{M+1}} \sum_{i=1}^N \ln \Pr \left(\mathbf{y}_i, \prod_{m \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_j^{(m)} \middle| \mathbf{x}_i, \gamma, \{\theta_m\}_{m=1}^{M+1} \right),$$

where latent variables are:

- $\mathbf{z}_i$ denoting the r.v. of expert selection,

10

Probabilistic L2D - Missing annotations

Objective

$$\max_{\gamma, \{\theta_m\}_{m=1}^{M+1}} \sum_{i=1}^N \ln \Pr \left(\mathbf{y}_i, \prod_{m \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(m)} \middle| \mathbf{x}_i, \gamma, \{\theta_m\}_{m=1}^{M+1} \right),$$

where latent variables are:

- $\mathbf{z}_i$ denoting the r.v. of expert selection,
- $\prod_{j \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(j)}$ denoting the r.v. of missing annotations.

10

Probabilistic L2D - Missing annotations

Parameter inference

Leverage the variational Expectation - Maximisation algorithm:

- *E-step*: approximate posterior $q(\mathbf{z}, \prod_{j \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(j)})$ via variational inference:

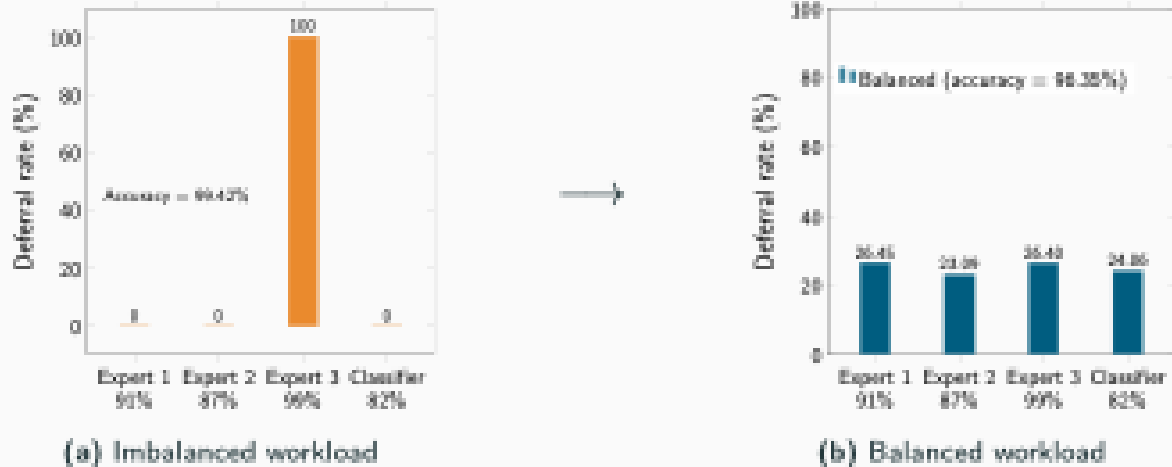
$$q \left(\mathbf{z}, \prod_{j \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(j)} \right) = q(\mathbf{z}) \prod_{j \in \mathcal{D}_i^{\text{obs.}}} q(\mathbf{t}_i^{(j)})$$

$$q^* = \underset{q}{\operatorname{argmin}} \operatorname{KL} \left[q \left(\mathbf{z}, \prod_{j \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(j)} \right) \middle| \Pr \left(\mathbf{z}_i, \prod_{j \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(j)} \middle| \mathbf{x}_i, \mathbf{y}_i, \prod_{m \in \mathcal{D}_i^{\text{obs.}}} \mathbf{t}_i^{(m)}, \gamma^{(k)}, \{\theta_m^{(k)}\}_{m=1}^{M+1} \right) \right]$$

- *M-step*: maximise the “complete”-data log-likelihood w.r.t. $\gamma, \{\theta_m\}_{m=1}^{M+1}$.

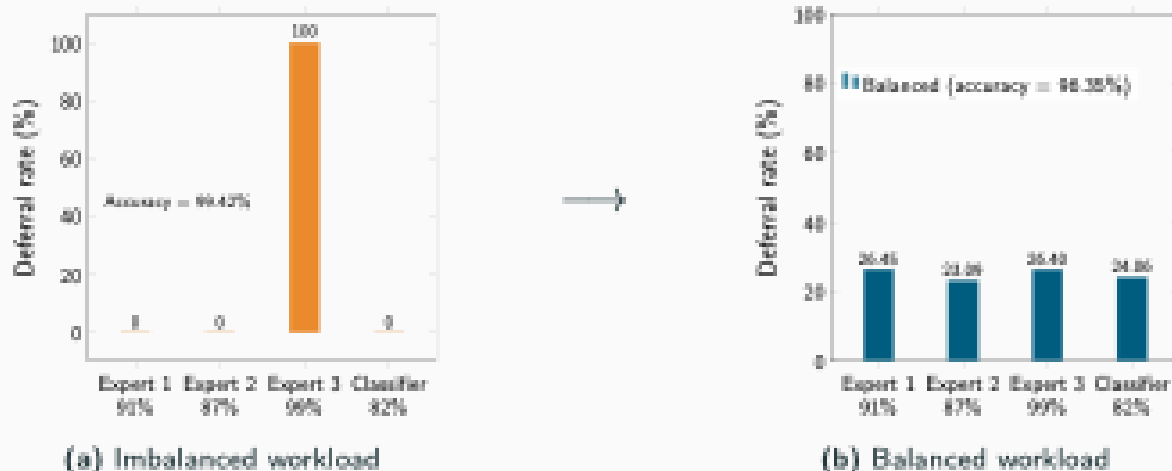
11

Probabilistic L2D - Control workload assignment



12

Probabilistic L2D - Control workload assignment



An additional E-step is introduced to calculate the constrained posterior:

$$\tilde{q}^* = \underset{\tilde{q}}{\operatorname{argmin}} \operatorname{KL} [\tilde{q}(\mathbf{z}) \| q^*(\mathbf{z})], \forall i \in \{1, \dots, N\} \quad \text{s.t.: } \epsilon_l \preceq \frac{1}{N} \sum_{i=1}^N \tilde{q}(\mathbf{z}) \preceq \epsilon_u,$$

where q^* and $\tilde{q}$ denote the unconstrained and constrained posteriors of $\mathbf{z}$.

12

Limitations

- *De-skilling* human by deferring one type of samples,

15

Limitations

- *De-skilling* human by deferring one type of samples,
- *Scalability* w.r.t. the number of human experts (current is $\mathcal{O}(n_{\text{human}})$)

15

Limitations

- *De-skilling* human by deferring one type of samples,
- *Scalability* w.r.t. the number of human experts (current is $\mathcal{O}(n_{\text{human}})$)
- *Static performance* assumption on human experts

15

Limitations

- *De-skilling* human by deferring one type of samples,
- *Scalability* w.r.t. the number of human experts (current is $\mathcal{O}(n_{\text{human}})$)
- *Static performance* assumption on human experts
- *Weak cooperation* between human and machine.

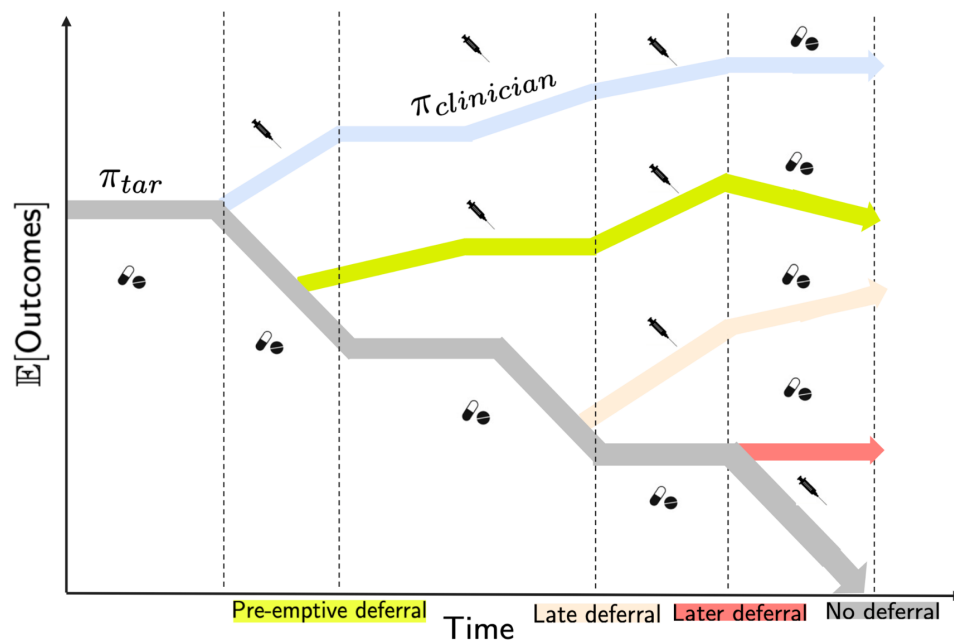
15

Future work

- “Reverse” L2D design for education purpose (e.g. train radiologists),
- To address the scalability issue:
 - use a shared model conditioned on “human representation”, or
 - cluster humans into groups
- Integrate dynamic performance of human (collaborate with psychologists and radiologists)
- Enhance further the human - machine cooperation.

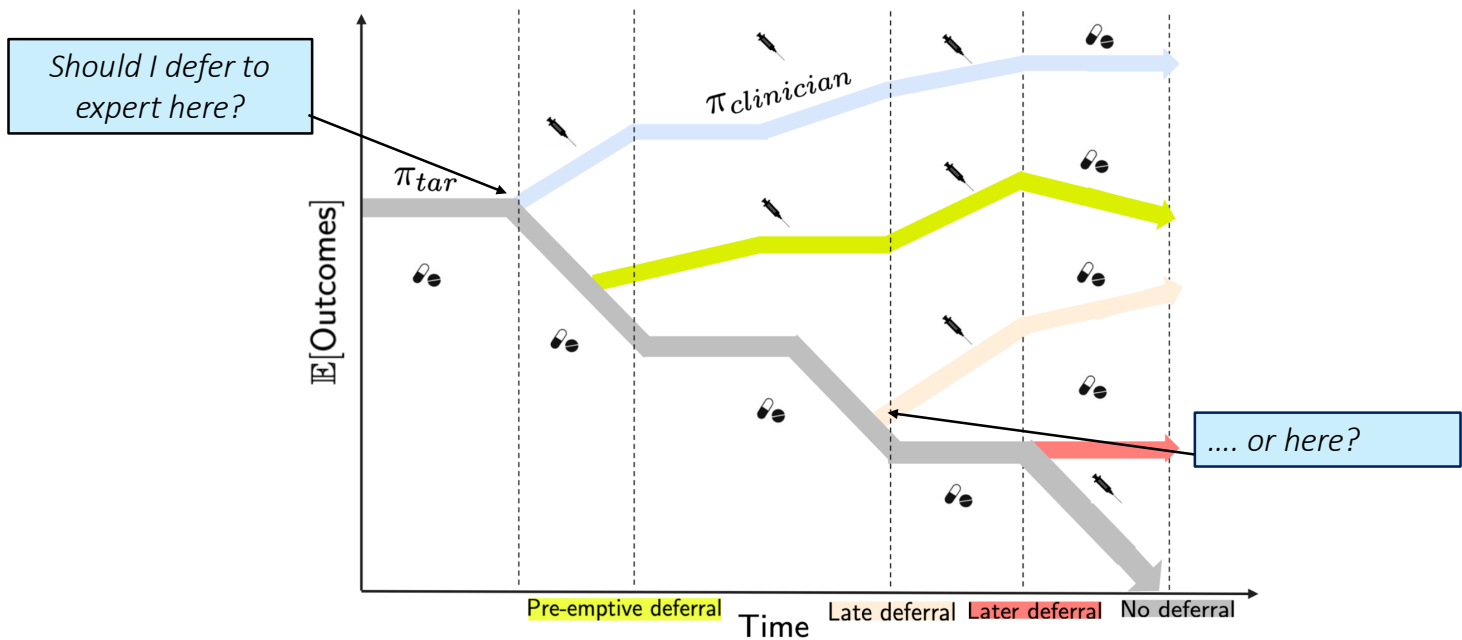
16

Bonus: Learning to Defer in Sequential Settings



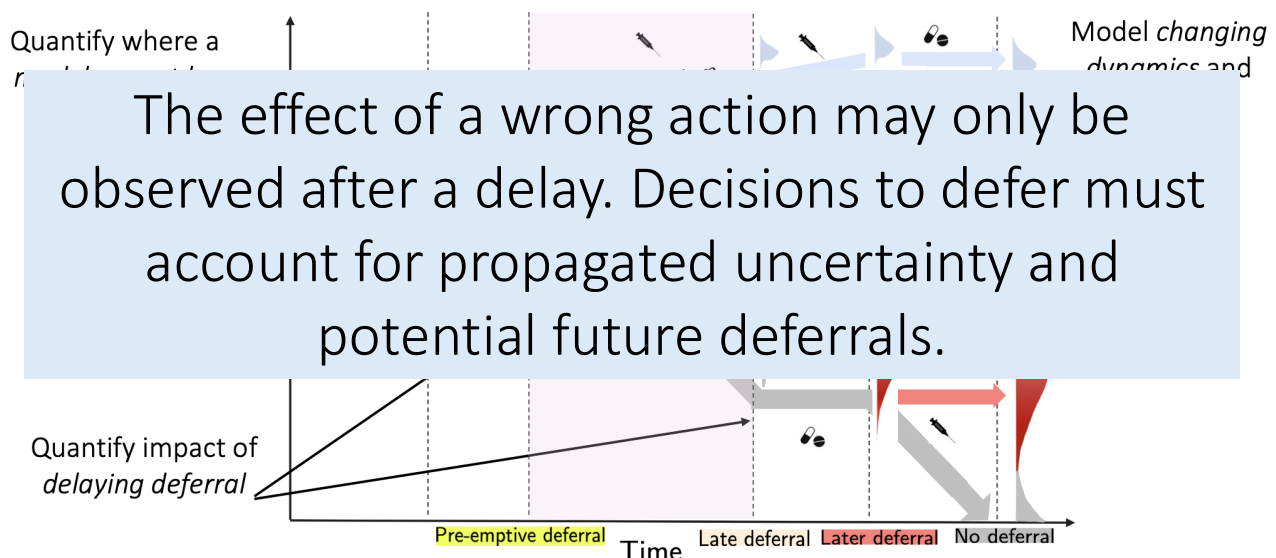
54

Bonus: Learning to Defer in Sequential Settings



55

Bonus: Learning to Defer in Sequential Settings



56

Bonus: Learning to Defer in Sequential Settings

Algorithm 1 Sequential Learning to Defer

Input: $\mathcal{D}^*$, behavior policy π_0 , target policy π_{tar} .
Estimate Posterior Distributions $\{\mathcal{M}_t \triangleq p_t(\cdot|\mathcal{D}^*)\}_{t=0}^T$ (posteriors over rewards not shown here)
Initialization: Deferral function $g_{\pi_{\text{tar}}}(s, t) = 0$ for all $s \in \mathcal{S}$ and $t \in \{1, 2, \dots, T\}$.
for $n \in \text{BOOTSTRAPS}(\mathcal{D}^*)$ **do**
 Sample $\mathcal{M}_k \triangleq \{\mathcal{M}_{k,t} \sim p_t(\cdot|\mathcal{D}^*)\}_{t \in \{1, 2, \dots, T\}, \forall k \in \{1, 2, \dots, K\}}$
 for $t \in \{T, T-1, \dots, 1\}$ **do**
 for $s \in \mathcal{S}$ **do**
 Compute $V_{\pi_{\text{tar}}(t), \text{mix}(t_+)}^{\mathcal{M}}, V_{\pi_0(t), \text{mix}(t_+)}^{\mathcal{M}} - c \forall \mathcal{M}$
 $\tilde{g}_{\pi_{\text{tar}}}(s, t) \leftarrow \approx \frac{1}{K} \sum_{\mathcal{M}_k \sim \{p_t(\cdot|\mathcal{D})\}_t} [\mathbf{1}(V_{\pi_{\text{tar}}(t), \text{mix}(t_+)}^{\mathcal{M}_k} < V_{\pi_0(t), \text{mix}(t_+)}^{\mathcal{M}_k} - c)]$
 end for
 end for
end for
return $g_{\pi_{\text{tar}}}(s, t) = \mathbf{1}(\tilde{g}_{\pi_{\text{tar}}}(s, t) > \tau) \forall s, t \in \mathcal{S} \times \{1, 2, \dots, T\}$

Joshi, S.*, Parbhoo, S. * and Doshi-Velez, F., 2023. Learning-to-defer for sequential medical decision-making under uncertainty. TMLR, <https://openreview.net/pdf?id=Opn3KnBH5F>.

57

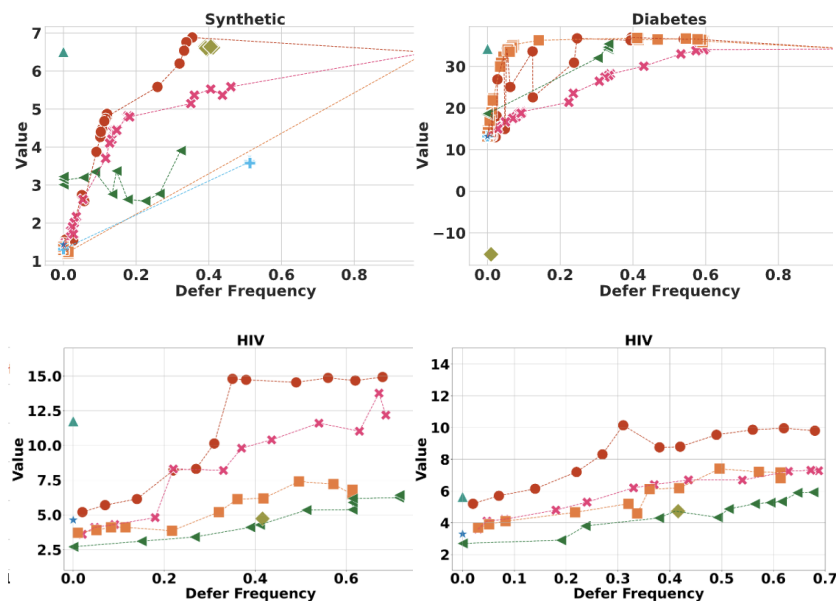
Our solution **outperforms** baselines

Method	Synthetic		Diabetes		HIV - Case Study I	
	Value (mean $\pm$ 2 s.e.)	Defer Frequency	Value (mean $\pm$ 2 s.e.)	Defer Frequency	Value (mean $\pm$ 2 s.e.)	Defer Frequency
SLTD	8.029 $\pm$ 0.039	0.509	36.931 $\pm$ 0.166	0.396	14.792 $\pm$ 0.267	0.342
SLTD-Stat.	5.588 $\pm$ 0.048	1.000	34.326 $\pm$ 0.172	1.000	11.020 $\pm$ 0.230	0.629
SLTD-One Step	5.578 $\pm$ 0.050	1.000	36.819 $\pm$ 0.23	0.412	9.671 $\pm$ 0.129	0.531
SLTD (K=1)	8.011 $\pm$ 0.025	0.511	36.678 $\pm$ 0.289	0.297	9.311 $\pm$ 0.162	0.517
SLTD-Stat. (K=1)	5.575 $\pm$ 0.036	1.000	34.320 $\pm$ 0.175	1.000	8.659 $\pm$ 0.027	0.571
SLTD-One Step (K=1)	5.595 $\pm$ 0.038	1.000	36.482 $\pm$ 0.284	0.328	8.640 $\pm$ 0.104	0.318
Augmented-MDP	3.044 $\pm$ 0.023	0.512	13.273 $\pm$ 0.126	0.000	N/A	N/A
Mozannar et. al.	6.369 $\pm$ 2.226	0.468	-15.155 $\pm$ 0.675	0.010	3.820 $\pm$ 0.231	0.291
Madras et. al.	5.731 $\pm$ 0.223	1.000	35.388 $\pm$ 0.475	0.332	3.330 $\pm$ 0.481	0.400
π_{tar}	-1.80 $\pm$ 0.071	N/A	13.202 $\pm$ 0.162	N/A	5.837 $\pm$ 0.171	N/A
π_0	5.485 $\pm$ 0.035	N/A	34.241 $\pm$ 0.151	N/A	14.124 $\pm$ 0.592	N/A

Joshi, S.*, Parbhoo, S. * and Doshi-Velez, F., 2023. Learning-to-defer for sequential medical decision-making under uncertainty. TMLR, <https://openreview.net/pdf?id=Opn3KnBH5F>.

58

Our solution **improves long-term outcomes with fewer deferrals**



Joshi, S.* , Parbhoo, S.* and Doshi-Velez, F., 2023. Learning-to-defer for sequential medical decision-making under uncertainty. TMLR, <https://openreview.net/pdf?id=Opn3KnH5F>.

59

Decomposing the uncertainty can help us **interpret when human expertise is beneficial**

High Epistemic Uncertainty

Possible improvement if additional data in this region could be collected to improve the reducible sources of uncertainty.

High Aleatoric Uncertainty

There is high variability in the patient's dynamics that may need to be managed with careful interventions or is otherwise not manageable.

	Defer Time t_d	Total Uncertainty	Modeling Uncertainty	Mean Outcome
Synthetic data	3	26.190	0.233	3.42
Diabetes	3	3418.17	34.160	73.669

50

Summary

- Bayesian models and conformal prediction give predictive uncertainty.
- Depending on how we communicate this uncertainty, it may be used in different ways.
- Learning to defer in uncertainty is a framework where we use uncertainty of experts to determine how to combine different opinions.
- Can be applied to classifiers/static tasks or dynamic settings.
- Probabilistic version of learning to defer avoids having complete annotations.